



# Deployment of Web Applications

**Siva R Jasthi**

Computer Sciences and Cybersecurity

Metropolitan State University

# Deployment of Web Applications

- Hosting Providers
- CPANEL
- Subdomains, Databases, FTP Accounts
- FileZilla
  - (Client to Server)
  - (Server to Client)

# Hosting Providers

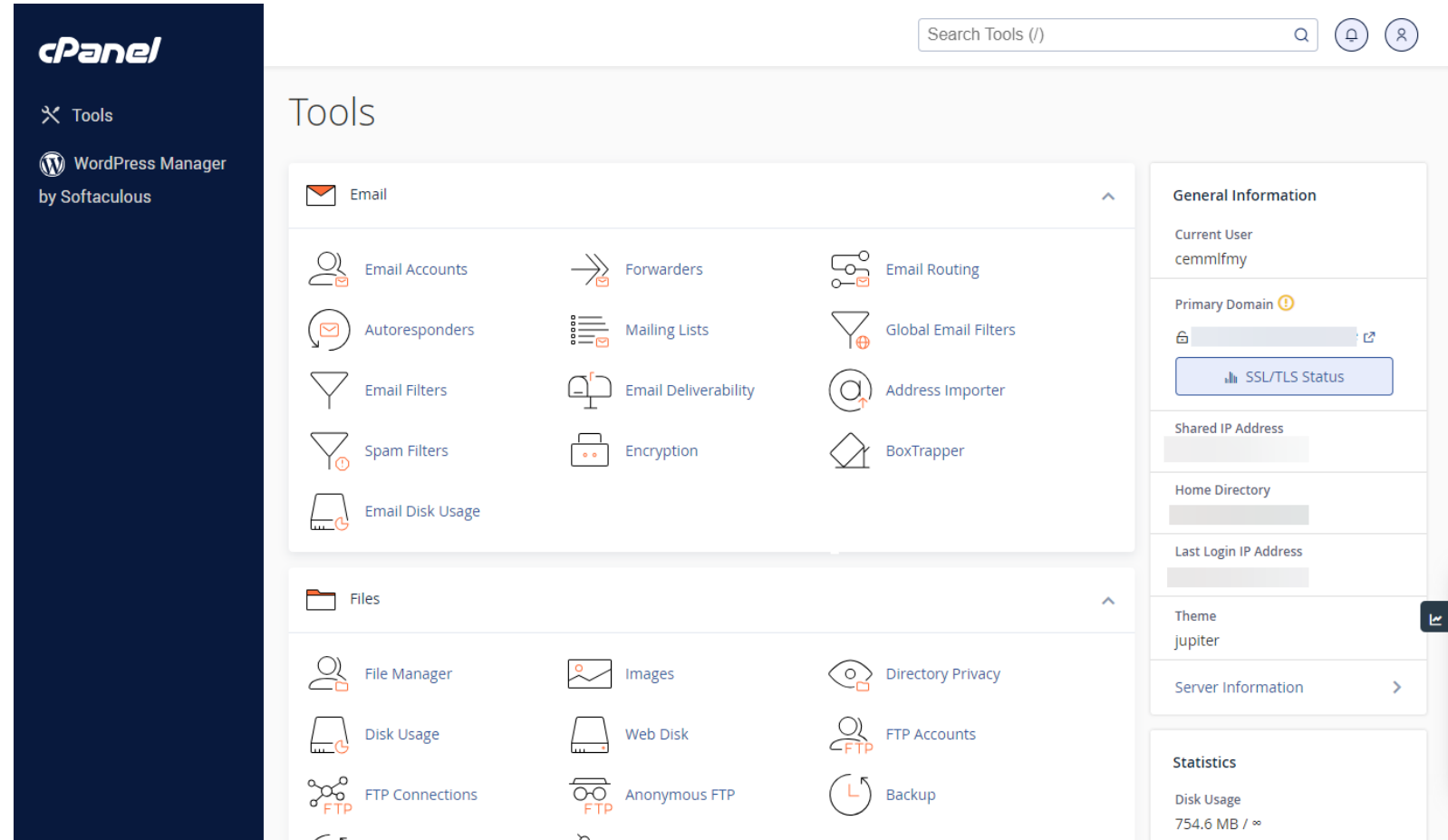


<https://www.top10.com/hosting/comparison>

- For example, all my projects are on bluehost

# What is cPanel?

- cPanel is a web-based control panel used by many hosting providers to manage website hosting accounts.
- It provides a graphical interface and automation tools designed to simplify the process of website management for both hosting providers and website owners.



# What is Cpanel? (Contd.)

- 1. File Management:** Users can upload, download, and manage files directly through the control panel using a File Manager.
- 2. Email Management:** cPanel allows users to create and manage email accounts associated with their domain, set up email forwarding, autoresponders, and spam filters.
- 3. Domain Management:** Users can add, remove, and manage domain names associated with their hosting account. They can also set up subdomains, addon domains, and parked domains.
- 4. Database Management:** cPanel provides tools for creating and managing MySQL and PostgreSQL databases, including phpMyAdmin for database administration.
- 5. Website Statistics:** Users can view various statistics about their website's performance, including traffic, visitor demographics, and bandwidth usage.
- 6. One-click Installs:** cPanel often includes a feature called Softaculous or Installatron, which allows users to quickly install popular web applications like WordPress, Joomla, and Drupal with just a few clicks.
- 7. Security Features:** cPanel offers various security features such as password-protected directories, SSL/TLS certificate management, IP address blocking, and more.

# What is Cpanel? (Contd.)



Email



Email Accounts



Forwarders



Email Routing



Autoresponders



Mailing Lists



Global Email Filters



Email Filters



Email Deliverability



Address Importer



Spam Filters



Encryption



BoxTrapper



Email Disk Usage

# What is Cpanel? (Contd.)



Files



File Manager



Images



Directory Privacy



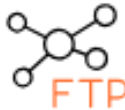
Disk Usage



Web Disk



FTP Accounts



FTP Connections



Anonymous FTP



Backup



Backup Wizard



Git™ Version Control

# What is Cpanel? (Contd.)



Databases



phpMyAdmin



MySQL® Databases



MySQL® Database Wizard



Remote MySQL®



PostgreSQL Databases



PostgreSQL Database Wizard



phpPgAdmin



# What is Cpanel? (Contd.)



Domains



Sitejet Builder



Domains



Redirects



Zone Editor

# What is Cpanel? (Contd.)



Metrics



Visitors



Site Quality Monitoring



Errors



Bandwidth



Raw Access



Awstats



Webalizer



Webalizer FTP



Metrics Editor

# What is Cpanel? (Contd.)



Security



[SSH Access](#)



IP Blocker



SSL/TLS



Manage API Tokens



Hotlink Protection



SSL/TLS Status

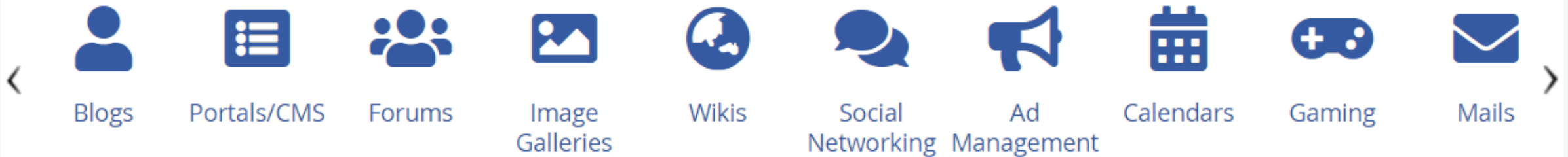
# What is Cpanel? (Contd.)

 Softaculous Apps Installer 

## Scripts:



## Categories :



# Subdomains

## Domains

List Domains

Use this interface to manage your domains. For more information, read the [documentation](#).



Enable Force HTTPS Redirect



| Domain ^                                                           | Document Root          | Redirects To   |
|--------------------------------------------------------------------|------------------------|----------------|
| <input type="checkbox"/> <a href="#">aalambana.jasthi.com</a>      | /public_html/aalambana | Not Redirected |
| <input type="checkbox"/> <a href="#">abcd2.projectabcd.com</a>     | /public_html/abcd2     | Not Redirected |
| <input type="checkbox"/> <a href="#">aeds.jasthi.com</a>           | /public_html/aeds      | Not Redirected |
| <input type="checkbox"/> <a href="#">ahala.telugupuzzles.com</a>   | /public_html/ahala     | Not Redirected |
| <input type="checkbox"/> <a href="#">animals.telugupuzzles.com</a> | /public_html/animals   | Not Redirected |
| <input type="checkbox"/> <a href="#">bingo.telugupuzzles.com</a>   | /public_html/bingo     | Not Redirected |

# FTP Accounts

- Each FTP Account has access to one of the public\_html folders on the remote host
- You need to use a file transfer client (such as FileZilla) to move the content from the client to server.
- Each project is provided with a FTP account (which will be active for the semester).
- The FTP account will be disabled at the end of the semester.

# Creation of FTP Accounts

## FTP Accounts

Add and configure FTP Accounts to get your website on the internet fast. You can use an FTP client to manage your website.

### Add FTP Account

Log In

Domain

Password

Password (Again)

Strength 

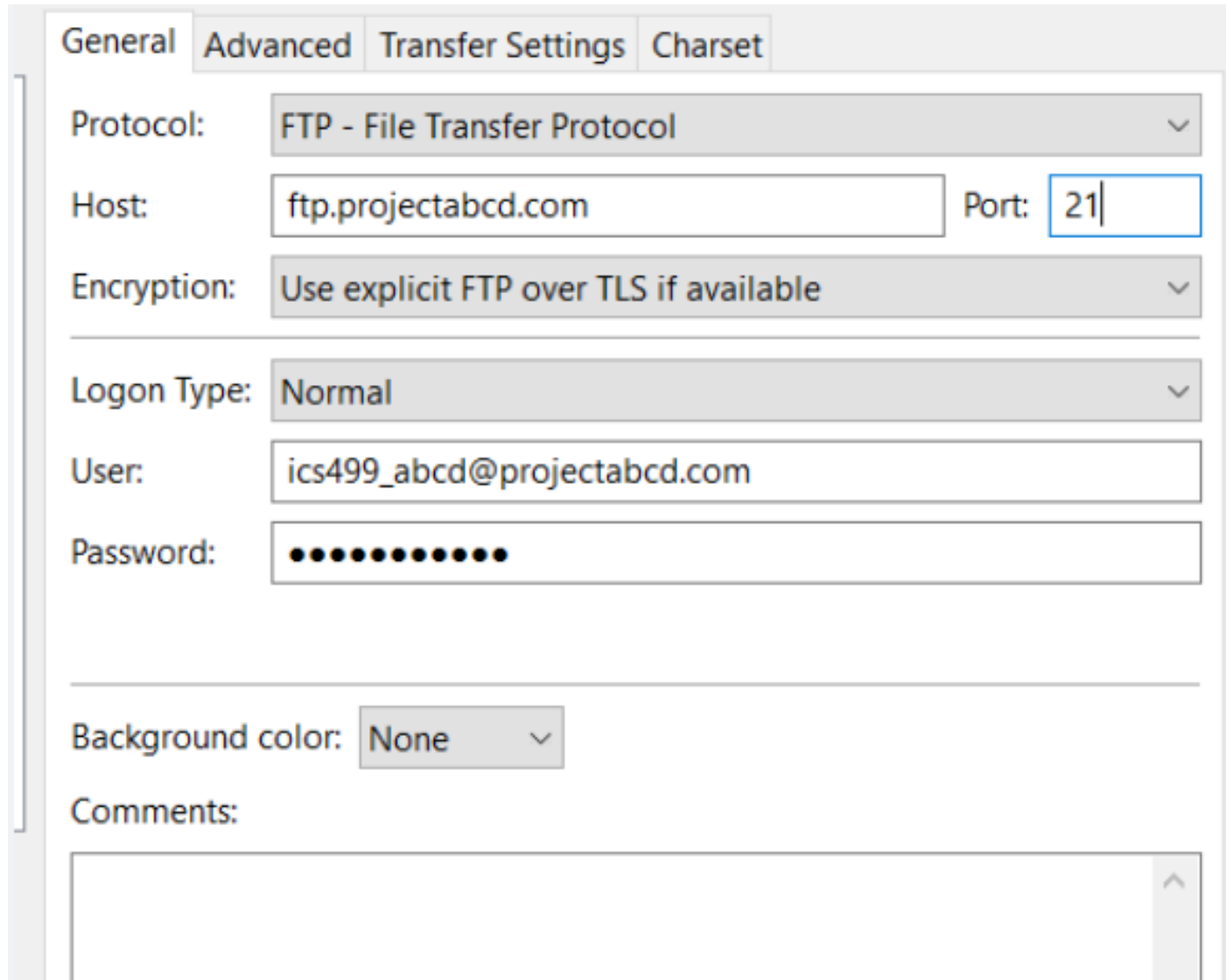
Very Weak (0/100)

Password Generator

Directory

/home2/icsbinco/

# Filezilla (Example 1)



The image shows the 'General' tab of the FileZilla settings window. The 'Protocol' is set to 'FTP - File Transfer Protocol'. The 'Host' is 'ftp.projectabcd.com' and the 'Port' is '21'. The 'Encryption' is set to 'Use explicit FTP over TLS if available'. The 'Logon Type' is 'Normal'. The 'User' is 'ics499\_abcd@projectabcd.com' and the 'Password' is masked with dots. The 'Background color' is set to 'None'. There is a 'Comments' text area at the bottom.

| General                                            | Advanced | Transfer Settings | Charset |
|----------------------------------------------------|----------|-------------------|---------|
| Protocol: FTP - File Transfer Protocol             |          |                   |         |
| Host: ftp.projectabcd.com                          |          | Port: 21          |         |
| Encryption: Use explicit FTP over TLS if available |          |                   |         |
| Logon Type: Normal                                 |          |                   |         |
| User: ics499_abcd@projectabcd.com                  |          |                   |         |
| Password: ●●●●●●●●                                 |          |                   |         |
| Background color: None                             |          |                   |         |
| Comments:                                          |          |                   |         |

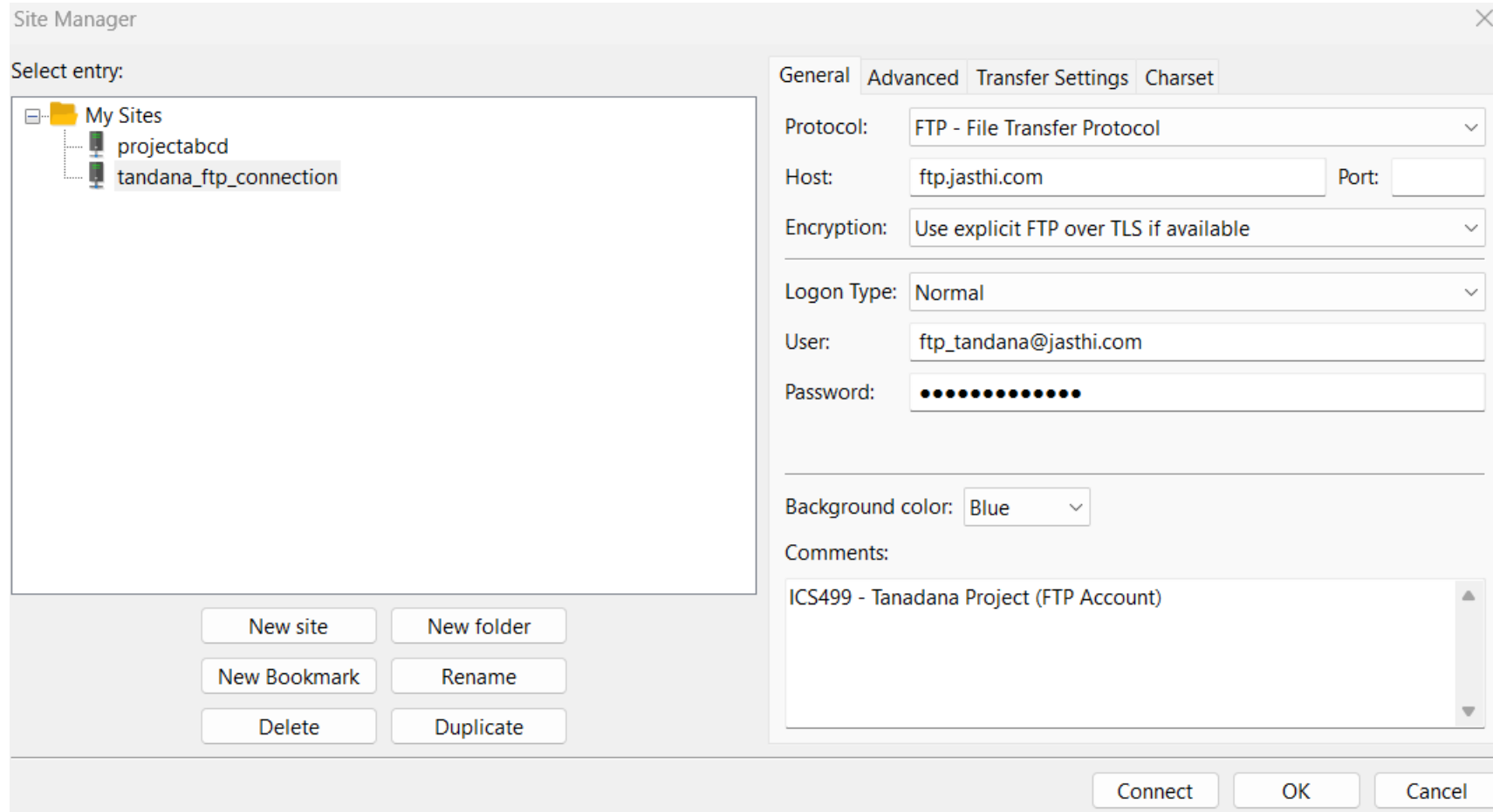
Team is promoting the code to GitHub

One of the team members will host the code to the remote host. They host the code on remote host.

And the DEMO happens on the remote host.



# Filezilla (You can save the configuration in FileZilla Client)



# Filezilla – File Transfer Considerations

- Any file you move from the client to the server will overwrite the existing file on the server.
- It is very very important to ensure that you do not cause regressions on the remote host. We should always be making progress.
- Do NOT host / move / transfer the `db_configuration.php` file. It will overwrite the connection information and your system will not work.

# Deployment Cheat Sheet

Onetime activity (to be done by Jasthi)

1. Create a subdomain
2. Identify a public\_html folder to hold the content
3. Create a database and import the data (eg: icsbinco\_tandana\_db)
4. Create a db user and assign the user to the database (eg: isbinco\_admin)
5. Set the configuration in a configuration file (change the db name, db user and password in the \*.php file)
6. Create FTP account and give access to the respective project/folder

Do this every week (by the Project Team)

1. Move the content from the client (desktop) to the remote host
2. Perform the testing on the remove host

# Create a subdomain

## Domains

[List Domains](#) / Create a New Domain

Use this interface to manage your domains. For more information, read the [documentation](#).

### Create a New Domain

#### Domain

Enter the domain that you would like to create:

tandana.jasthi.com

#### Document Root (File System Location)

If the document root is shared then the created domain will serve the same content as "jasthi.com". **This setting is permanent.**

☐ Share document root (/home2/icsbinco/public\_html) with "jasthi.com".



tandana

Specify the directory where you want the files for this domain to exist.



Submit And Create Another

[Return To Domains](#)

# Assigning a user to a database with privileges

## MySQL® Databases

### Manage User Privileges

User: **icsbinco\_tandana\_admin**

Database: **icsbinco\_tandana\_db**

☒ ALL PRIVILEGES

☒ ALTER

☒ ALTER ROUTINE

☒ CREATE

☒ CREATE ROUTINE

☒ CREATE TEMPORARY TABLES

☒ CREATE VIEW

☒ DELETE

☒ DROP

# Deployment Checklist (Bluehost)

## Checklist for Deployment:

### Team Info:

Team Name: Dolphins

Team Membership:

DevOps Engineer:

### Project Info:

GitHub Link (upstream):

Trello Link:

### Localhost Links:

Domain/URL:

Database Name:

Database User Name:

Database User Password:

### BlueHost Links (to be provided by instructor)

Domain/URL: version4.projectabcd.com

Database Name: icsbinco\_version4abcd\_db

Database User Name: icsbinco\_version4\_user

Database User Password: ICS325\_version4

FTP User Name: version4@jasthi.com

FTP User Password: ICS325\_version4

FTP home directory: public\_html/version4

### Deployment Instructions:

Received the instructions from the instructor: ( ) yes ( ) not yet

Team able to deploy changes using FileZilla: ( ) yes ( ) not yet

# Summary

1. Create a subdomain
2. Identify a public\_html folder to hold the content
3. Create a database and import the data
4. Set the configuration in a configuration file
5. Create FTP account and give access to the respective project/folder
6. Move the content from the client (desktop) to the remote host
7. Perform the testing on the remove host

**Q&A**

